

Algorithm 2: Floating-Point Changes to Equations (5) and (7)

Notation and Arithmetic Model

Let $A \in \mathbb{R}^{m \times n}$. The i -th row of A is A_{i*} , and

$$D(A) = \|A\|_F^2 = \sum_{i=1}^m \|A_{i*}\|_2^2.$$

Assume $D(A) > 0$. Algorithm 2 samples indices

$$I_1, \dots, I_s \in \{1, \dots, m\}$$

independently from the exact row-norm probability law

$$p_i = \frac{\|A_{i*}\|_2^2}{D(A)}.$$

The probability symbol $\mathbb{P}[\cdot]$ is always taken with respect to this exact product sampling law. This is the row-sampling law from Algorithm 2 in Drineas and Mahoney's RandNLA survey [1].

The exact sampled sketch is $\tilde{A} \in \mathbb{R}^{s \times n}$, with entries

$$\tilde{A}_{tj} = \frac{A_{I_t j}}{\sqrt{s p_{I_t}}}.$$

The exact Gram matrix of a matrix B is $B^T B$. Thus the exact sampled Gram matrix is $\tilde{A}^T \tilde{A}$, and the exact reference Gram matrix is $A^T A$.

The floating-point unit roundoff is $u \geq 0$. When a Gram entry is computed by a length- s floating-point dot product, γ_s denotes the usual valid dot-product error factor. The probabilities p_i and the sampling law are exact by convention, but the non-probability denominator

$$d_i = \sqrt{s p_i}$$

is computed. The concrete denominator routine formalized here is

$$\hat{d}_i = \text{flsqrt}(\text{flmul}(s, p_i)).$$

It satisfies, for every positive-probability row,

$$|\hat{d}_i - d_i| \leq d_i \alpha(u), \quad \alpha(u) = \sqrt{1+u} u + u, \quad \hat{d}_i \neq 0.$$

Define the support relative denominator budget and the resulting effective row-scaling relative error by

$$\rho_A = \sum_{i: p_i > 0} \frac{d_i \alpha(u)}{|\hat{d}_i|}, \quad \eta_A = u + (1+u)\rho_A.$$

The deterministic perturbation budgets used below are

$$\begin{aligned}\tau_{\text{scale}}^{\text{comp}}(A) &= n(2\eta_A + \eta_A^2)D(A), \\ \tau_{\text{dot}}^{\text{comp}}(A; s) &= n\gamma_s(1 + \eta_A)^2D(A),\end{aligned}$$

and

$$\tau_{\text{full}}^{\text{comp}}(A; s) = \tau_{\text{scale}}^{\text{comp}}(A) + \tau_{\text{dot}}^{\text{comp}}(A; s).$$

For comparison, if d_i is supplied exactly, then $\rho_A = 0$, $\eta_A = u$, and these reduce to the older exact-denominator budgets.

What Is Exact and What Is Computed

- Exact arithmetic inputs: the matrix A , the probabilities p_i , the random samples I_t , the exact sketch $\tilde{A}$, and the exact reference Gram matrix $A^T A$.
- Computed denominator table: $\hat{d}_i$ is computed by $\text{flsqrt}(\text{flmul}(s, p_i))$. This is a floating-point non-probability quantity and its error contributes through ρ_A and η_A .
- Floating-point row-scaling output: $\hat{A}_{\text{comp}} \in \mathbb{R}^{s \times n}$ denotes the sketch whose entries are produced as $\text{fldiv}(A_{I_t j}, \hat{d}_{I_t})$.
- Exact arithmetic after a floating-point sketch: the matrix $\hat{A}_{\text{comp}}^T \hat{A}_{\text{comp}}$ is the exact mathematical Gram matrix of the already rounded sketch. No dot-product kernel is charged in this model.
- Floating-point computed Gram output: $\hat{G}_{\text{dot}}(A)$ denotes the Gram matrix whose entries are computed by floating-point dot products applied to the computed-denominator rounded sketch $\hat{A}_{\text{comp}}$. This model charges denominator computation, rounded row scaling, and dot products through $\tau_{\text{full}}^{\text{comp}}(A; s)$.
- Equation (7) exact analysis basis: the matrix U is the exact orthonormal-column basis used to define the leverage probabilities, the exact sampling law, and the reference matrix I_n .
- Equation (7) computed basis table: the final stored-basis theorem below uses a rounded table $\hat{U}^{(\mu)}$ in the sampled sketch and charges its entrywise storage error, the computed denominator table, rounded row divisions, and rounded Gram dot products. The three closed storage modes are $\mu \in \{\text{mul1}, \text{add0}, \text{sub0}\}$.
- Equation (7) actual-input endpoint: when an exact analysis factorization $A = UC$ is supplied with $U^T U = I$, the theorem below samples rows of the actual input A using the exact leverage law defined by U . The computed non-probability quantities are the concrete denominator table, the rounded row divisions on entries of A , and the rounded Gram dot products. The factors U and C are exact analysis/probability witnesses in that theorem, not computed algorithm outputs.
- Equation (7) basis generation from a data matrix: the theorem below does not claim that a QR, SVD, or rank-revealing routine has computed U from a matrix A . Such a routine must provide its own computed-basis theorem before it can be composed with this stored-basis endpoint.

Equation (5)

Exact baseline (Equation (5), exact arithmetic). *This is equation (5) in the RandNLA survey [1]. For $s > 0$ and $\varepsilon > 0$,*

$$\mathbb{P}\left[\left\|\tilde{A}^T \tilde{A} - A^T A\right\|_F \leq \varepsilon D(A)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Theorem 1 (Equation (5), computed denominators and rounded row scaling). *For $s > 0$ and $\varepsilon > 0$,*

$$\mathbb{P}\left[\left\|\hat{A}_{\text{comp}}^T \hat{A}_{\text{comp}} - A^T A\right\|_F \leq \varepsilon D(A) + \tau_{\text{scale}}^{\text{comp}}(A)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

Thus the exact equation (5) sampling probability is unchanged, while the threshold is enlarged by the proved deterministic denominator-plus-division budget.

Theorem 2 (Equation (5), fully computed denominators and Gram). *Assume the length- s dot-product factor γ_s is valid. For $s > 0$ and $\varepsilon > 0$,*

$$\mathbb{P}\left[\left\|\hat{G}_{\text{dot}}(A) - A^T A\right\|_F \leq \varepsilon D(A) + \tau_{\text{full}}^{\text{comp}}(A; s)\right] \geq 1 - \frac{1}{s\varepsilon^2}.$$

The additive floating-point part has order

$$\tau_{\text{full}}^{\text{comp}}(A; s) = \Theta(nD(A)(\eta_A + \eta_A^2 + \gamma_s(1 + \eta_A)^2)).$$

In the small-roundoff regime where $\eta_A \rightarrow 0$ and $\gamma_s \rightarrow 0$,

$$\tau_{\text{full}}^{\text{comp}}(A; s) = \Theta(nD(A)(\eta_A + \gamma_s)).$$

For the concrete denominator routine above, $\alpha(u) = \Theta(u)$, and η_A is determined by u , the exact probabilities, and the computed denominators $\hat{d}_i$. For fixed A, m, n, s and nonzero computed positive-probability denominators, the additive FP budget tends to zero as $u, \gamma_s \rightarrow 0$, so the result is non-vacuous.

Equation (7)

Now let $U \in \mathbb{R}^{m \times n}$ have orthonormal columns:

$$U^T U = I_n.$$

Assume $n > 1$. Algorithm 2 samples rows from the exact leverage-score law

$$p_i^U = \frac{\|U_{i*}\|_2^2}{n}.$$

Let $\tilde{U} \in \mathbb{R}^{s \times n}$ be the exact leverage-sampled sketch. For symmetric matrices $X, Y \in \mathbb{R}^{n \times n}$, write $X \preceq Y$ when

$$x^T X x \leq x^T Y x \quad \text{for every } x \in \mathbb{R}^n.$$

Exact baseline (Equation (7), exact Bennett form). *This is the leverage-score subspace-embedding bound corresponding to equation (7) in the RandNLA survey [1]. The underlying leverage-sampling theorem appears, for example, in Drineas–Mahoney–Muthukrishnan–Sarlos [2] and is stated here*

through the rank-one covariance concentration route of Oliveira [3] and Tropp-style matrix concentration [4]. Assume $s > 0$, $\varepsilon > 0$, $\delta > 0$, and

$$\log \frac{2n}{\delta} \leq s \frac{\varepsilon^2}{2(n-1) + (2/3)n\varepsilon}, \quad \log \frac{2n}{\delta} \leq s \frac{\varepsilon^2}{2(n-1) + (2/3)\varepsilon}.$$

Then

$$\mathbb{P}[\tilde{U}^T \tilde{U} - I_n \preceq \varepsilon I_n \text{ and } I_n - \tilde{U}^T \tilde{U} \preceq \varepsilon I_n] \geq 1 - \delta.$$

Theorem 3 (Equation (7), computed-denominator floating-point Bennett form). *Assume the hypotheses and sample-budget inequalities in the exact Bennett baseline, and assume the length- s dot-product factor γ_s is valid. Let $\hat{G}_{\text{dot}}(U)$ be the floating-point computed Gram matrix formed from the leverage-sampled sketch using the concrete computed denominator $\hat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U))$, rounded row scaling, and rounded length- s Gram dot products. Then*

$$\mathbb{P}[\hat{G}_{\text{dot}}(U) - I_n \preceq (\varepsilon + \tau_{\text{full}}^{\text{comp}}(U; s))I_n \text{ and } I_n - \hat{G}_{\text{dot}}(U) \preceq (\varepsilon + \tau_{\text{full}}^{\text{comp}}(U; s))I_n] \geq 1 - \delta.$$

Since $U^T U = I_n$, $D(U) = \|U\|_F^2 = n$, and therefore

$$\tau_{\text{full}}^{\text{comp}}(U; s) = n^2(2\eta_U + \eta_U^2 + \gamma_s(1 + \eta_U)^2).$$

Its small-roundoff order is

$$\tau_{\text{full}}^{\text{comp}}(U; s) = \Theta(n^2(\eta_U + \gamma_s)), \quad \eta_U = u + (1 + u) \sum_{i: p_i^U > 0} \frac{\sqrt{sp_i^U} \alpha(u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|}.$$

Actual Input Matrix for Equation (7)

This subsection states the end-to-end row-sampling conclusion for the actual matrix whose leverage scores are analyzed. To avoid overloading dimensions, let $U \in \mathbb{R}^{m \times r}$ have orthonormal columns, let $C \in \mathbb{R}^{r \times q}$, and define

$$A = UC \in \mathbb{R}^{m \times q}.$$

The exact leverage probabilities are

$$p_i^U = \frac{\|U_{i*}\|_2^2}{r}.$$

The factors U and C are exact analysis/probability witnesses here: the theorem does not assert that a QR, SVD, or rank-revealing routine has computed them. The implemented sampled sketch uses rows of A , not rows of U . Its exact row t is

$$\tilde{A}_{U,t*} = \frac{A_{I_t*}}{\sqrt{sp_{I_t}^U}}, \quad I_t \sim p^U.$$

Assume $r > 1$, $s > 0$, $\varepsilon > 0$, $\delta > 0$, and

$$\log \frac{2r}{\delta} \leq s \frac{\varepsilon^2}{2(r-1) + (2/3)r\varepsilon}, \quad \log \frac{2r}{\delta} \leq s \frac{\varepsilon^2}{2(r-1) + (2/3)\varepsilon}.$$

Then exact arithmetic gives the relative Gram statement

$$\mathbb{P}[\tilde{A}_U^T \tilde{A}_U - A^T A \preceq \varepsilon A^T A \text{ and } A^T A - \tilde{A}_U^T \tilde{A}_U \preceq \varepsilon A^T A] \geq 1 - \delta.$$

For floating-point arithmetic, the concrete leverage denominator is

$$\widehat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U)),$$

and the effective row-scaling relative error is

$$\eta_U^A = u + (1 + u) \sum_{i: p_i^U > 0} \frac{\sqrt{s p_i^U} (\sqrt{1 + u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|}.$$

For each actual-input column j , define

$$L_j^A = \sum_{i: p_i^U > 0} \frac{|A_{ij}|}{\sqrt{p_i^U}}.$$

The fully computed actual-input radius is

$$T_A^{\text{fp}} = (\gamma_s(1 + \eta_U^A)^2 + 2\eta_U^A + (\eta_U^A)^2) \sum_{j=1}^q (L_j^A)^2.$$

With all local shorthand expanded, this is

$$\begin{aligned} T_A^{\text{fp}} = & \left[\gamma_s \left(1 + u + (1 + u) \sum_{i: \|U_{i*}\|_2^2/r > 0} \frac{\sqrt{s \|U_{i*}\|_2^2/r} (\sqrt{1 + u} u + u)}{|\text{flsqrt}(\text{flmul}(s, \|U_{i*}\|_2^2/r))|} \right)^2 \right. \\ & + 2 \left(u + (1 + u) \sum_{i: \|U_{i*}\|_2^2/r > 0} \frac{\sqrt{s \|U_{i*}\|_2^2/r} (\sqrt{1 + u} u + u)}{|\text{flsqrt}(\text{flmul}(s, \|U_{i*}\|_2^2/r))|} \right) \\ & \left. + \left(u + (1 + u) \sum_{i: \|U_{i*}\|_2^2/r > 0} \frac{\sqrt{s \|U_{i*}\|_2^2/r} (\sqrt{1 + u} u + u)}{|\text{flsqrt}(\text{flmul}(s, \|U_{i*}\|_2^2/r))|} \right)^2 \right] \\ & \cdot \sum_{j=1}^q \left(\sum_{i: \|U_{i*}\|_2^2/r > 0} \frac{|\sum_{\ell=1}^r U_{i\ell} C_{\ell j}|}{\sqrt{\|U_{i*}\|_2^2/r}} \right)^2. \end{aligned}$$

Theorem 4 (Equation (7), actual input matrix and computed Gram). *Under the hypotheses above, assume the length- s dot-product factor γ_s is valid. Let $\widehat{G}_{\text{dot}}(A; U)$ be the floating-point Gram computed by sampling exact rows of $A = UC$ from the exact leverage law defined by U , computing $\widehat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U))$, rounding the sampled-row divisions, and computing the Gram entries by floating-point dot products. Then*

$$\mathbb{P} \left[\widehat{G}_{\text{dot}}(A; U) - A^T A \preceq \varepsilon A^T A + T_A^{\text{fp}} I_q \text{ and } A^T A - \widehat{G}_{\text{dot}}(A; U) \preceq \varepsilon A^T A + T_A^{\text{fp}} I_q \right] \geq 1 - \delta.$$

Remark. Let $k_U = |\{i : p_i^U > 0\}|$. In the fixed-problem small-roundoff regime where U, C, m, r, q, s are fixed, positive-probability computed denominators remain nonzero and converge to their exact values, and $u, \gamma_s \rightarrow 0$,

$$\eta_U^A = \Theta(u(1 + k_U)).$$

Consequently

$$T_A^{\text{fp}} = \Theta \left((\gamma_s + \eta_U^A) \sum_{j=1}^q (L_j^A)^2 \right) = \Theta \left((\gamma_s + u(1 + k_U)) \sum_{j=1}^q \left(\sum_{i: p_i^U > 0} \frac{|A_{ij}|}{\sqrt{p_i^U}} \right)^2 \right).$$

The radius tends to zero in this regime, so the computed actual-input theorem is non-vacuous. If a concrete implementation first computes U or C , that QR/SVD/rank-revealing routine must be certified separately before this exact-analysis-witness theorem is claimed for that generation path.

Stored Basis Table for Equation (7)

The basis-sampled computed-denominator theorem above charges the computed denominator, row divisions, and dot products while using the exact entries of U in the sampled sketch. The next result is the implementation-facing stored-basis endpoint for the modeled rounded-copy paths. The exact basis U still defines the probability law and the reference identity matrix, but the computed sketch uses a stored table.

Let

$$\mathcal{M} = \{\text{mul1}, \text{add0}, \text{sub0}\}.$$

For $\mu \in \mathcal{M}$, define the stored basis table by

$$\widehat{U}_{ij}^{(\text{mul1})} = \text{flmul}(U_{ij}, 1), \quad \widehat{U}_{ij}^{(\text{add0})} = \text{fladd}(U_{ij}, 0), \quad \widehat{U}_{ij}^{(\text{sub0})} = \text{flsub}(U_{ij}, 0).$$

For each of these three storage modes,

$$|\widehat{U}_{ij}^{(\mu)} - U_{ij}| \leq u|U_{ij}|.$$

The leverage denominator is still computed by

$$d_i^U = \sqrt{s p_i^U}, \quad \widehat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U)).$$

On positive-probability rows, the concrete denominator routine proves

$$|\widehat{d}_i^U - d_i^U| \leq d_i^U \alpha(u), \quad \widehat{d}_i^U \neq 0.$$

Define the denominator relative budget

$$\rho_U = \sum_{i: p_i^U > 0} \frac{\sqrt{s p_i^U} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|}.$$

The total relative sampled-entry budget from denominator formation, rounded division, and one rounded-copy storage of the basis table is

$$\lambda_U = 2u + u^2 + (1+u)^2 \rho_U.$$

For each column j , define the exact positive-support column magnitude

$$B_j^U = \sum_{i: p_i^U > 0} \frac{|U_{ij}|}{\sqrt{p_i^U}}.$$

The stored-basis sampled-Gram perturbation radius is

$$T_U^{\text{fp}} = (\gamma_s(1 + \lambda_U)^2 + 2\lambda_U + \lambda_U^2) \sum_{j=1}^n (B_j^U)^2.$$

Equivalently, with every shorthand replaced,

$$\begin{aligned} T_U^{\text{fp}} = & \left[\gamma_s \left(1 + 2u + u^2 + (1 + u)^2 \sum_{i: p_i^U > 0} \frac{\sqrt{sp_i^U} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|} \right)^2 \right. \\ & + 2 \left(2u + u^2 + (1 + u)^2 \sum_{i: p_i^U > 0} \frac{\sqrt{sp_i^U} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|} \right) \\ & \left. + \left(2u + u^2 + (1 + u)^2 \sum_{i: p_i^U > 0} \frac{\sqrt{sp_i^U} (\sqrt{1+u} u + u)}{|\text{flsqrt}(\text{flmul}(s, p_i^U))|} \right)^2 \right] \sum_{j=1}^n \left(\sum_{i: p_i^U > 0} \frac{|U_{ij}|}{\sqrt{p_i^U}} \right)^2. \end{aligned}$$

Theorem 5 (Equation (7), stored basis table and computed Gram). *Assume the hypotheses and sample-budget inequalities in the exact Bennett baseline, and assume the length- s dot-product factor γ_s is valid. Fix any storage mode $\mu \in \mathcal{M}$. Let $\widehat{G}_{\text{dot}}^{(\mu)}(U)$ be the matrix whose entries are computed from the exact leverage samples by using the stored table $\widehat{U}^{(\mu)}$, the concrete denominator table $\widehat{d}_i^U = \text{flsqrt}(\text{flmul}(s, p_i^U))$, rounded row divisions, and rounded length- s Gram dot products. Then*

$$\mathbb{P} \left[\widehat{G}_{\text{dot}}^{(\mu)}(U) - I_n \preceq (\varepsilon + T_U^{\text{fp}}) I_n \text{ and } I_n - \widehat{G}_{\text{dot}}^{(\mu)}(U) \preceq (\varepsilon + T_U^{\text{fp}}) I_n \right] \geq 1 - \delta.$$

This is a final theorem for the three modeled stored-table paths $\text{flmul}(U_{ij}, 1)$, $\text{fladd}(U_{ij}, 0)$, and $\text{flsub}(U_{ij}, 0)$. It is not a theorem about computing U from a data matrix by QR, SVD, or another basis-generation routine.

Remark. In the small-roundoff regime with fixed U, m, n, s , nonzero computed positive-probability denominators, and $\widehat{d}_i^U/d_i^U \rightarrow 1$, the denominator sum satisfies

$$\rho_U = \Theta(u |\{i : p_i^U > 0\}|), \quad \lambda_U = \Theta(u(1 + |\{i : p_i^U > 0\}|)).$$

Thus the stored-basis radius has the compact order

$$T_U^{\text{fp}} = \Theta \left((\gamma_s + \lambda_U) \sum_{j=1}^n (B_j^U)^2 \right),$$

or, with the column magnitude expanded,

$$T_U^{\text{fp}} = \Theta \left((\gamma_s + u(1 + |\{i : p_i^U > 0\}|)) \sum_{j=1}^n \left(\sum_{i: p_i^U > 0} \frac{|U_{ij}|}{\sqrt{p_i^U}} \right)^2 \right).$$

Since $U^T U = I_n$ and $n > 1$, the displayed column factor is finite and positive. Therefore $T_U^{\text{fp}} \rightarrow 0$ as $u, \gamma_s \rightarrow 0$ in this fixed problem regime, so the stored-basis theorem is non-vacuous.

Remark. The older vector-action consequence has probability

$$1 - \frac{1}{s(\varepsilon/n)^2}$$

and the exact-denominator threshold obtained by setting $\rho_U = 0$ and $\eta_U = u$. The Bennett finite-Loewner results above are the current Algorithm 2 theorem surfaces for the concrete computed-denominator path and for the three modeled stored-basis paths.

References

References

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